2.8.1 Sum to n terms of an A.P.

A series whose terms are in Arithmetic progression is called Arithmetic series.

Let $a, a+d, a+2d, a+3d, \ldots$ be the Arithmetic Progression.

The sum of first n terms of a Arithmetic Progression denoted by S_n is given by,

$$S_n = a + (a+d) + (a+2d) + \cdots + (a+(n-1)d) \qquad \ldots(1)$$

Rewriting the above in reverse order

$$S_n = (a+(n-1)d) + (a+(n-2)d) + \cdots + (a+d) + a \qquad \ldots(2)$$

Adding (1) and (2) we get,

$$2S_n = [a+a+(n-1)d] + [a+d+a+(n-2)d] + \cdots + [a+(n-2)d+(a+d)] + [a+(n-1)d+a]$$

$$= [2a+(n-1)d] + [2a+(n-1)d + \cdots + [2a+(n-1)d] \quad (n \text{ terms})$$

$$2S_n = n \times [2a+(n-1)d] \quad \Rightarrow \quad S_n = \frac{n}{2}\left[2a+\left(n-1\right)d\right]$$

Note

If the first term a, and the last term l (n^{th} term) are given then

$$S_n = \frac{n}{2}\left[2a+\left(n-1\right)d\right] = \frac{n}{2}\left[a+a+\left(n-1\right)d\right] \quad (\because\ l = a+\left(n-1\right)d)$$

$$S_n = \frac{n}{2}[a+l].$$

Progress Check

1. The sum of terms of a sequence is called _____.
2. If a series have finite number of terms then it is called _____.
3. A series whose terms are in _____ is called Arithmetic series.
4. If the first and last terms of an A.P. are given, then the formula to find the sum is _____.

Example 2.31 Find the sum of first 15 terms of the A. P. $8, 7\frac{1}{4}, 6\frac{1}{2}, 5\frac{3}{4}, \ldots$

Solution Here the first term $a = 8$, common difference $d = 7\frac{1}{4} - 8 = -\frac{3}{4}$,

Sum of first n terms of an A.P. $S_n = \frac{n}{2}\left[2a+(n-1)d\right]$

$$S_{15} = \frac{15}{2}\left[2\times 8 + (15-1)(-\frac{3}{4})\right]$$

$$S_{15} = \frac{15}{2}\left[16 - \frac{21}{2}\right] = \frac{165}{4}$$

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